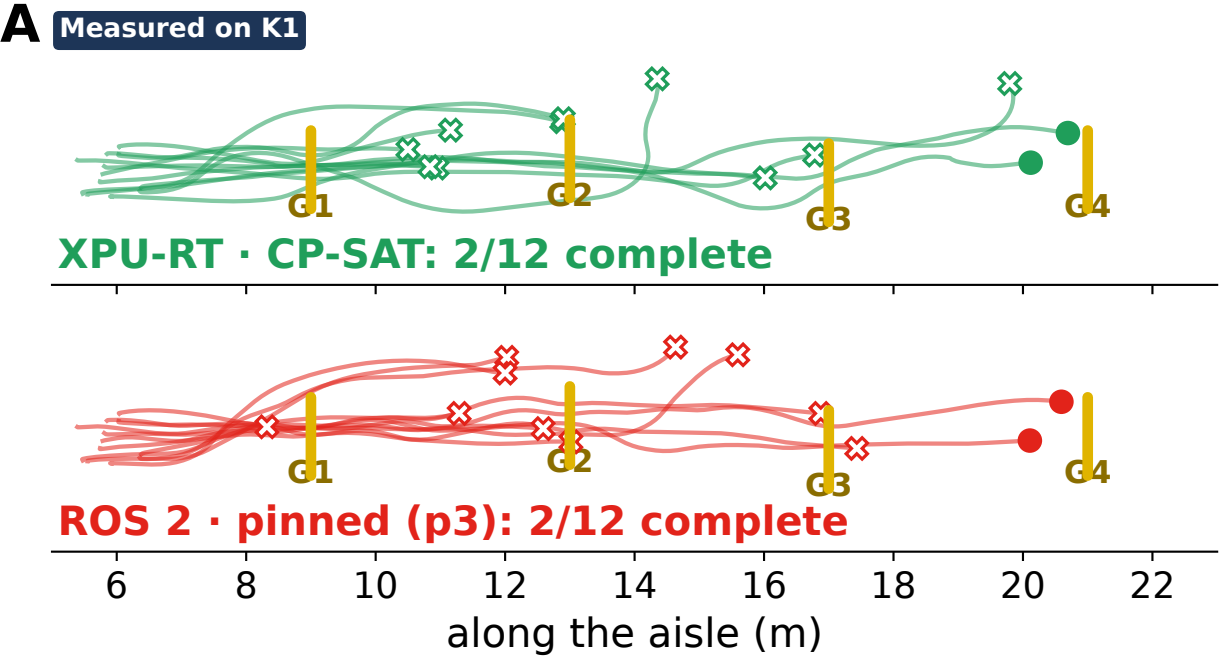


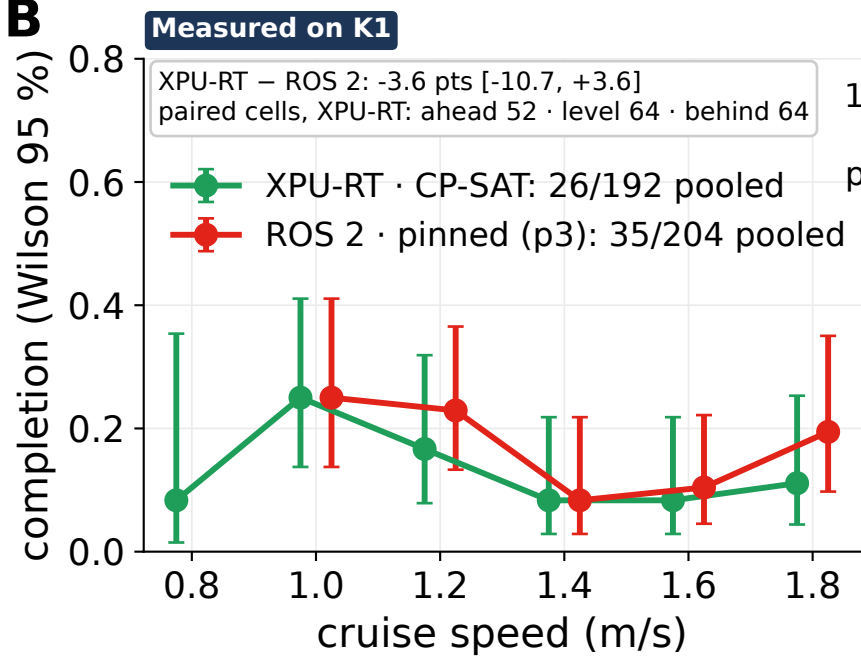
Warehouse showdown, ROS 2 arm re-costed from the board's own traces: the arm the calibrated model describes (pinned, control on its own timer) runs 100 Hz control on the K1 and flies level with XPU-RT

No flight replays a model cadence: A-B fly the measured cadences of `ros\_p345.csv` (the arm tier B predicts within 2.1 ms camera→goal and 0.0 Hz control rate) and `xpu\_a\_cpsat\_hard.csv` in the simulator. Census: XPU-RT 26/192, ROS 2 35/204 complete — a tie, not a win.

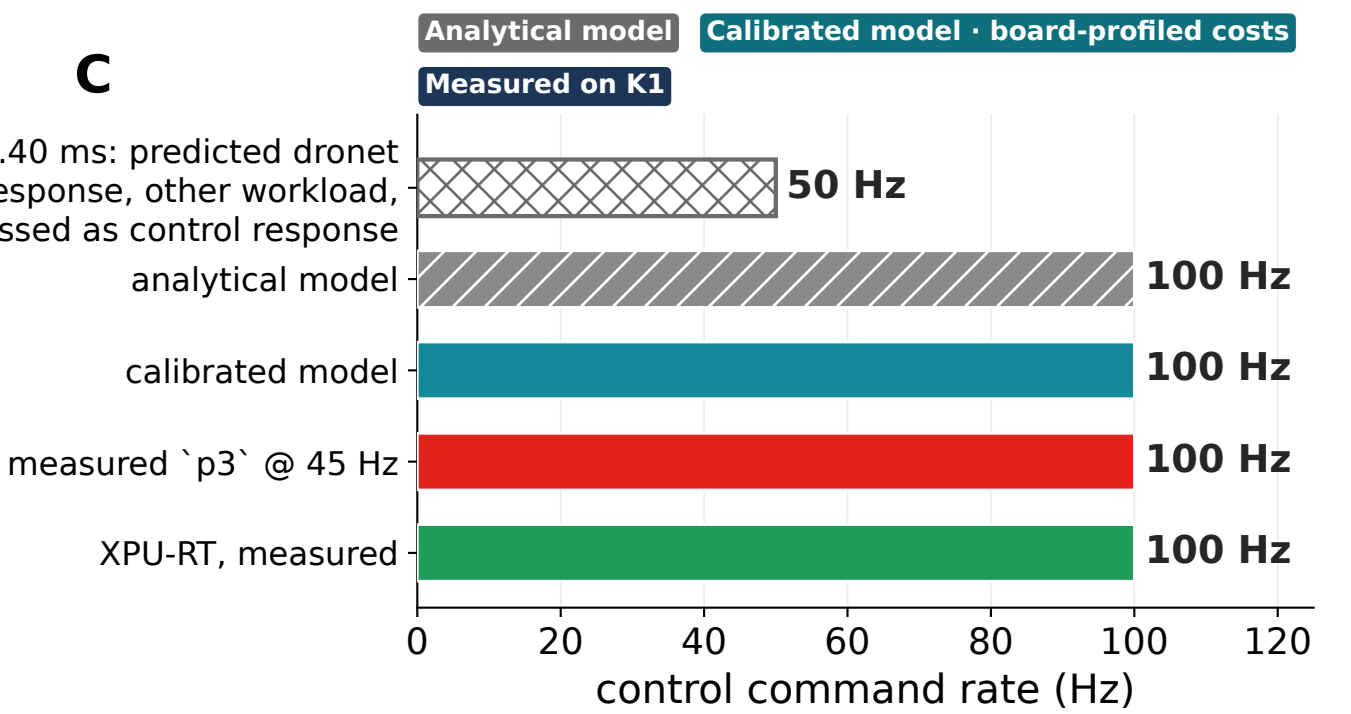
Every census flight of one cell (1.0 m/s, density 0.30): ● complete, × crash



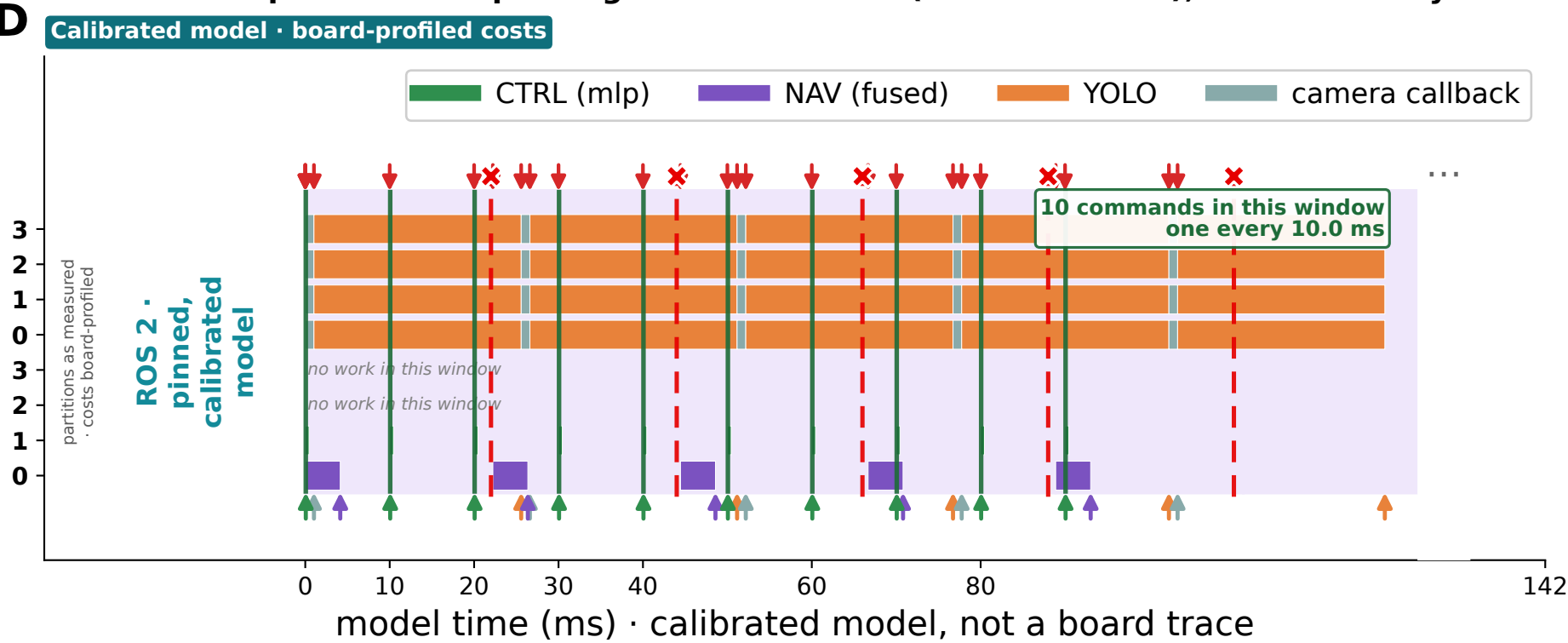
B Census: all speeds and densities



Control rate: printed · predicted · measured



Calibrated model, pinned arm, 45 Hz camera: pin_timeline on board-profiled costs camera→goal 28.7 + queue 25.6 = 54.3 ms vs 56.4 measured · control 100 vs 100.0 Hz without a drop rule the response grows each frame (25.6 → 39.0 ms); the board stays flat



Measured on the K1, 45 Hz camera, one 100 ms window of each arm's board trace XPU-RT · CP-SAT: camera→control 57 ms, control every 10.0 ms ROS 2 · pinned (p3): camera→control 61 ms, control every 10.0 ms, 2/772 frames late

